The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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Abstract

We identify a fundamental structural asymmetry between General Relativity (GR) and Quantum Field Theory (QFT): quantum fields require a fixed kinematic background to define locality and operator algebra, whereas GR asserts that the background metric is itself a dynamical physical degree of freedom. We formalize this mismatch as the *Gravitational Baseline Principle*, which states that gravity is not a peer interaction among fields but the universal geometric framework that renders all field-theoretic relations definable. Treating gravity as a conventional quantizable gauge field leads to contradictions in background independence and gauge covariance. To resolve this, we propose a minimal, background-independent quantum substrate from which both smooth manifolds and quantum field algebras co-emerge. Geometry arises from the entanglement structure of a relational density matrix, and the graviton appears as a collective, low-energy excitation of this substrate. We show how the Laplace–Beltrami operator, Einstein dynamics, and thermodynamic gravitational laws emerge in the continuum limit. This framework provides a conceptually coherent path toward unifying GR and QFT without quantizing the metric as a fundamental field.

1 Introduction

Quantum Field Theory (QFT) describes matter and radiation as operator-valued distributions defined on a fixed background spacetime [2]. Locality, microcausality, and the construction of a Fock space all require a predetermined metric structure. General Relativity (GR), by contrast, denies the existence of any prior geometric container: the metric itself is a dynamical field determined by the Einstein equations [3]. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity [1].

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and the very notion of locality. Gauge interactions, by contrast, act only on specific internal symmetry representations. This motivates the *Gravitational Baseline Principle*: gravity is the universal geometric framework that renders all other interactions definable. It is not a peer field within a background but the background itself.

In this paper, we develop a minimal pre-geometric quantum substrate from which smooth geometry and quantum fields co-emerge. The substrate is defined by a relational density matrix whose entanglement structure encodes emergent spatial connectivity. We show how the Laplace–Beltrami operator, graviton excitations, and thermodynamic gravitational dynamics arise in the continuum limit.

2 The Structural Asymmetry Problem

Einstein's field equations,

$$G_{\mu\nu} = 8\pi G \, T_{\mu\nu}, \tag{1}$$

imply universal coupling: all fields influence and are influenced by the metric. Gauge interactions, however, operate only on specific internal degrees of freedom.

Perturbative quantization of gravity requires expanding the metric as $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and quantizing $h_{\mu\nu}$ as a spin-2 gauge field. But this procedure presupposes a fixed background $\eta_{\mu\nu}$, violating diffeomorphism invariance at the foundational level. This is the core structural asymmetry: QFT requires a background; GR denies one.

3 The Gravitational Baseline Principle

Gravity is not a selective interaction but the universal geometric baseline. All physical systems couple non-linearly to the metric, which defines causal structure and measurement. The Planck scales,

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}}, \tag{2}$$

mark the breakdown of smooth geometry and the onset of pre-geometric structure.

The Gravitational Baseline Principle states:

> *Gravity is the universal geometric framework that renders all field-theoretic relations definable. It cannot be quantized as a conventional gauge field without violating background independence.*

4 Pre-Geometric Substrate Architecture

We define the pre-geometric substrate $\mathcal{Q}$ as a background-independent quantum system with state vector $|\Psi\rangle$ in a Hilbert space $\mathcal{H}_{\mathcal{Q}}$. The density matrix $\rho = |\Psi\rangle\langle\Psi|$ encodes relational structure.

Mutual information between subsystems i and j,

$$I(i,j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{3}$$

defines a symmetric connectivity matrix A_{ij}.

4.1 Emergent Continuum Geometry

A discrete structural operator acts on test functions $\phi(x)$ mapped to subsystem nodes:

$$\Delta_{\mathcal{Q}} f_i = \sum_j A_{ij}(f_i - f_j). \tag{4}$$

Expanding $\phi(x)$ in a Taylor series and imposing isotropy constraints yields the continuum Laplace–Beltrami operator:

$$\lim_{\alpha \to 0} \Delta_{\mathcal{Q}} = \nabla^2. \tag{5}$$

4.2 Dynamical Law

We propose a minimal background-independent evolution equation:

$$\frac{d\rho}{dt} = -i[H_{\mathcal{Q}}, \rho], \tag{6}$$

with Hamiltonian

$$H_{\mathcal{Q}} = \sum_{i,j} K(A_{ij})\, \sigma_i \otimes \sigma_j, \tag{7}$$

where K is a monotonic kernel of entanglement strength. Because A_{ij} depends on ρ, the evolution is self-consistent and non-linear.

5 Emergent Graviton Dynamics

Small perturbations of the emergent metric,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \tag{8}$$

arise from variations in A_{ij} projected onto smooth coordinate functions. After projection onto the transverse-traceless sector, the effective action reduces to the Pauli–Fierz form, demonstrating that the graviton is a collective excitation of the substrate.

6 Thermodynamic Gravity

Following Jacobson [4], Einstein's equations emerge as a thermodynamic equation of state. Fluctuations of the substrate generate higher-curvature corrections:

$$S_{\text{eff}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left(R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \cdots \right), \tag{9}$$

with coefficients determined by the entanglement spectrum.

7 Phenomenological Consequences

- **Holographic noise:** Mutual-information encoding implies an irreducible noise floor in precision interferometry.
- **Curvature corrections:** Higher-order terms may affect black hole quasi-normal modes and early-universe cosmology.
- **Modified dispersion:** Collective excitations satisfy $\omega^2 = k^2[1 + \epsilon(k)]$, with $\epsilon(k)$ suppressed by the microscopic scale.

8 Relation to Existing Approaches

Unlike Loop Quantum Gravity or Causal Set Theory, which quantize classical geometric structures, this framework treats geometry as emergent from non-spatial quantum information. It shares conceptual ground with ER=EPR [5] and entanglement-induced geometry [6], but provides a concrete dynamical substrate.

9 Limitations and Outlook

This framework is conceptual and minimalistic. A full treatment requires:

- explicit reconstruction of Lorentzian signature,
- derivation of matter field algebras,
- analysis of renormalization behavior,
- numerical simulations of substrate dynamics.

Nonetheless, the model provides a coherent path toward unifying GR and QFT without quantizing the metric.

10 Conclusion

The Gravitational Baseline Principle resolves the structural asymmetry between GR and QFT by recognizing gravity as the universal geometric framework rather than a conventional gauge field. A pre-geometric quantum substrate with relational entanglement structure yields emergent geometry, graviton excitations, and thermodynamic gravitational dynamics. This approach offers a conceptually consistent foundation for quantum gravity.

References

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